TAI LIEU ON TAP MON THIET KE PHAN MEM (SWD)

Mon hoc: Software Architecture & Design (SWD)

Tong quan kien truc va cac mau thiet ke cot loi gom 3 phan sau:

1. Kien truc Ba lop (Three-Tier Architecture):

- Presentation Tier: Giao dien tuong tac nguoi dung, nhan input va hien thi ket qua.
- Business Logic Tier: Xu ly nghiep vu cot loi, kiem tra logic va tinh toan du lieu.
- Data Tier: Co so du lieu, luu tru va truy van du lieu an toan.
- Uu diem: Phan tach trach nhiem (Separation of Concerns), giup he thong de bao tri va mo rong.

2. Mau thiet ke Singleton Pattern (Creational Pattern):

- Muc dich: Dam bao mot lop chi co duy nhat mot the hien (instance) trong toan bo ung dung.
- Co che: Constructor private, bien static giu instance va phuong thuc GetInstance().
- Ung dung: Quan ly ket noi Database Connection Pool, Logger tap trung, Configuration Manager.

3. Mau thiet ke Observer Pattern (Behavioral Pattern):

- Muc dich: Dinh nghia moi quan he mot-nhieu (one-to-many) giua cac doi tuong.
- Co che: Khi Subject thay doi trang thai, tat ca Observer dang ky se duoc tu dong thong bao.
- Ung dung: He thong xu ly su kien (Event-driven), Gui thong bao real-time (Notifications, Webhooks).